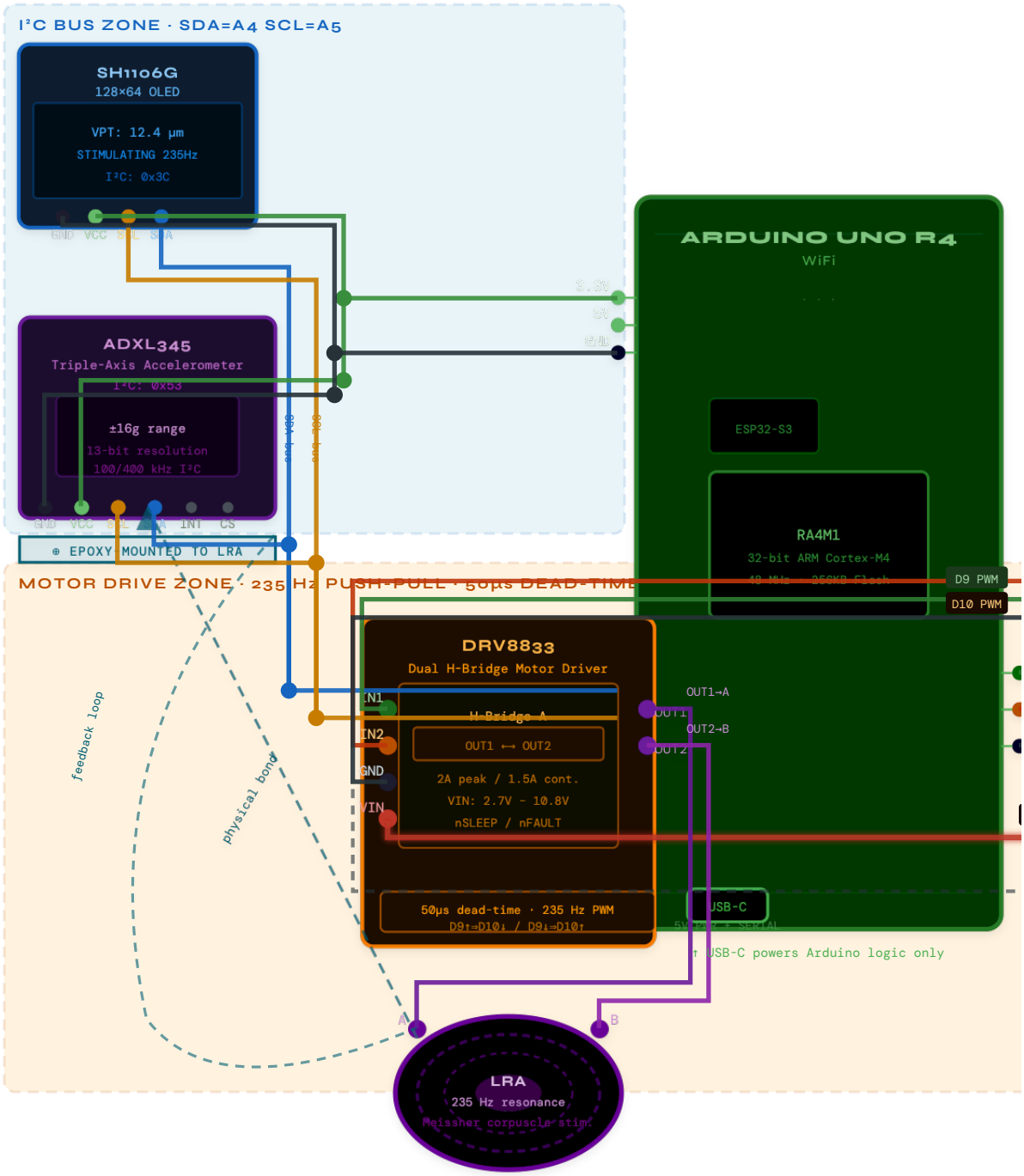
VPT DIAGNOSTIC SYSTEM — WIRING DIAGRAM

VIBRATION PERCEPTION THRESHOLD · 235 Hz
CLOSED-LOOP · REV 1.0



VPT DIAGNOSTIC — WIRING DIAGRAM

Arduino Uno R4 WiFi · DRV8833 H-Bridge · LRA @ 235 Hz · ADXL345 (closed-loop) · SH1106G OLED · External DC Supply

WIRE LEGEND

EXT DC V IN (High-current power rail)

GND (Common ground star) 3.3V (Arduino regulated → sensors)

I²C SDA (A4) I²C SCL (A5) D9 PWM → DRV8833 IN1

D10 PWM → DRV8833 IN2 LRA Drive (OUT1/OUT2 → A/B)

Closed-loop feedback path (physical + I²C)

Common GND tie (dashed = reference connection)